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SYNOPSIS Presentation

on

Military Stress Analysis

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PROJECT OVERVIEW

Stress is one of the most critical psychological and physiological challenges in modern society. With increasing academic pressure, corporate workload, military responsibilities, healthcare duties, and fast-paced urban lifestyles, individuals are constantly exposed to mental strain. Prolonged or unmanaged stress can lead to severe consequences such as anxiety disorders, depression, cardiovascular diseases, burnout, sleep disturbances, and reduced cognitive performance. In high-risk professions such as soldiers, doctors, nurses, air traffic controllers, and long-distance drivers, stress can not only affect personal health but also endanger public safety.

Traditional stress detection approaches rely on self-report questionnaires or clinical assessments, which are subjective, time-consuming, and often unreliable. Many automated systems have attempted to detect stress using either facial expressions or voice analysis alone. However, single-modality systems suffer from significant limitations. Facial expressions can be consciously manipulated, and a person may hide their stress. Similarly, voice-based analysis alone may be influenced by environmental noise, language differences, or speaking style. Due to these weaknesses, single-modality systems often produce inconsistent or inaccurate results.

To overcome these limitations, this project proposes a multimodal approach that integrates both voice and facial cues for stress screening. By combining information from multiple sources, the system can capture both visible emotional signals (facial movements, micro-expressions, eye behavior) and acoustic patterns (pitch variation, speech energy, tone, and prosody). This fusion of modalities improves reliability, robustness, and accuracy compared to traditional methods.

It is important to clarify that the proposed system is designed strictly as a screening and monitoring tool, not as a medical diagnostic system. It does not replace psychologists, psychiatrists, or professional mental health evaluations. Instead, it serves as an early-warning mechanism that can encourage users to seek help or adopt stress-management techniques when needed.

PROJECT OBJECTIVES

The primary objective of this research is to design and implement an intelligent, real-time system capable of assessing human stress levels using both voice and facial data. The system aims to bridge the gap between psychological theory and practical technology by applying machine learning techniques to behavioral signals.

First, the system will collect real-time audio and video data through a standard webcam and microphone. These signals will then be processed to extract meaningful features related to emotional and physiological states. From facial data, features such as action units, head pose, eye gaze, blink rate, and micro-expressions will be captured. From speech data, features including pitch, intensity, Mel-frequency cepstral coefficients (MFCC), and prosodic patterns will be extracted.

Second, these multimodal features will be fused into a unified representation that captures both visual and acoustic stress indicators. A machine learning model such as Random Forest, Support Vector Machine (SVM), or a Neural Network will be trained to classify stress into three levels: Low, Moderate, and High.

Third, the project will include the development of a user-friendly dashboard that visually displays the predicted stress level along with a confidence score. This dashboard will be simple, intuitive, and suitable for non-technical users such as students, teachers, or corporate employees.

Finally, the proposed approach will be validated using a publicly available research dataset, specifically the DAIC-WOZ corpus, ensuring scientific credibility and reproducibility of results.

PROJECT SCOPE

The scope of this project is broad yet clearly defined. The system will operate in real time using commonly available hardware such as a laptop, webcam, and microphone, making it practical, scalable, and low-cost for large-scale deployment. Unlike expensive clinical tools, this system can be easily installed in schools, colleges, offices, hospitals, and training centers without requiring specialized equipment or technical expertise.

In educational institutions, the system can help teachers and counselors identify students who may be experiencing excessive academic stress during exams, presentations, or competitive assessments. In corporate environments, it can be integrated into wellness programs to monitor employee burnout, workload pressure, and mental fatigue, enabling organizations to promote healthier work culture. In military and defense settings, it can assist in monitoring soldiers' psychological well-being during training or high-risk missions. Similarly, in healthcare, it can be used to track stress levels of doctors, nurses, and paramedics who frequently work under extreme pressure.

The system is particularly useful for early stress screening and awareness. It can help individuals recognize stress patterns before they become severe or lead to mental health disorders such as anxiety, depression, or chronic burnout. Researchers can also use the system for behavioral studies related to mental health, emotion recognition, human-computer interaction, and affective computing. It can serve as a valuable tool for psychology, neuroscience, and AI-based mental health research.

However, the system has clear ethical and practical boundaries. It is **not** intended to replace mental health professionals or provide medical diagnoses. The results should be considered only as preliminary indicators of stress, not as definitive conclusions. Additionally, the system will strictly respect user privacy by **not storing personal data, video recordings, or voice samples without explicit user consent**. Data security, encryption, and anonymity will be prioritized to prevent misuse of sensitive information.

METHODOLOGY

The technical methodology of the project follows a structured six-step approach.

Step 1: Data Collection

The project will utilize the DAIC-WOZ dataset, which is widely used in affective computing research. This dataset contains synchronized audio, video, and psychological labels collected during clinical-style interviews.

Step 2: Feature Extraction

Facial features will be extracted using OpenFace, which provides detailed analysis of facial action units, head movements, eye gaze direction, and subtle expressions. Voice features will be extracted using openSMILE, capturing acoustic properties such as pitch, energy, speech rate, and spectral characteristics.

Step 3: Data Preprocessing

Raw features will be cleaned to remove missing or corrupted values. Data will be normalized to ensure consistency, and audio-video timestamps will be synchronized for accurate fusion.

Step 4: Model Training

A supervised machine learning model will be trained to classify stress into Low, Moderate, and High categories based on labeled data.

Step 5: Real-Time System

In the real-time prototype, live video and audio will be captured, processed, and fed into the trained model for instant stress prediction.

Step 6: Evaluation

Model performance will be evaluated using accuracy, precision, recall, F1-score, and confusion matrix.

SYSTEM ARCHITECTURE

The overall system follows a modular and scalable pipeline that ensures flexibility, efficiency, and real-time performance. The process begins with data acquisition through a standard **webcam and microphone**, which continuously capture live video and speech signals from the user. These raw inputs are then processed through two specialized feature extraction modules.

For visual analysis, **OpenFace** is used to extract detailed facial features such as action units, head pose, eye gaze direction, blink rate, and micro-expressions that are strongly correlated with emotional and stress-related behaviors. Simultaneously, for audio analysis, **openSMILE** extracts acoustic and speech-based features including pitch variation, speech intensity, Mel-frequency cepstral coefficients (MFCC), prosody, and voice energy patterns that reflect psychological states.

The extracted features from both modalities are then combined using a **feature fusion technique**, which integrates facial and voice information into a unified dataset. This fused feature vector is passed to a trained **machine learning model** such as Random Forest, SVM, or Neural Network, which classifies the user's stress level into Low, Moderate, or High categories.

Finally, the results are displayed on a **Stress Dashboard**, which provides a clear visual representation of the predicted stress level along with a confidence score, making the system user-friendly and suitable for real-world applications.



HARDWARE REQUIREMENTS

The proposed system is designed to function efficiently using commonly available hardware, making it cost-effective and accessible for institutions and individuals. A standard laptop or desktop computer is sufficient to run the system. The device must include a functional **webcam** for capturing facial expressions and a **microphone** for recording speech signals in real time.

For smooth performance, a minimum of **8GB RAM** is recommended to handle video processing, audio feature extraction, and machine learning computations simultaneously. An **Intel i5 processor (or equivalent)** is suggested to ensure real-time responsiveness without lag or delay. Optional but beneficial hardware includes an external high-quality microphone for clearer voice capture and an HD webcam for better facial feature detection, which can further improve system accuracy.

SOFTWARE REQUIREMENTS

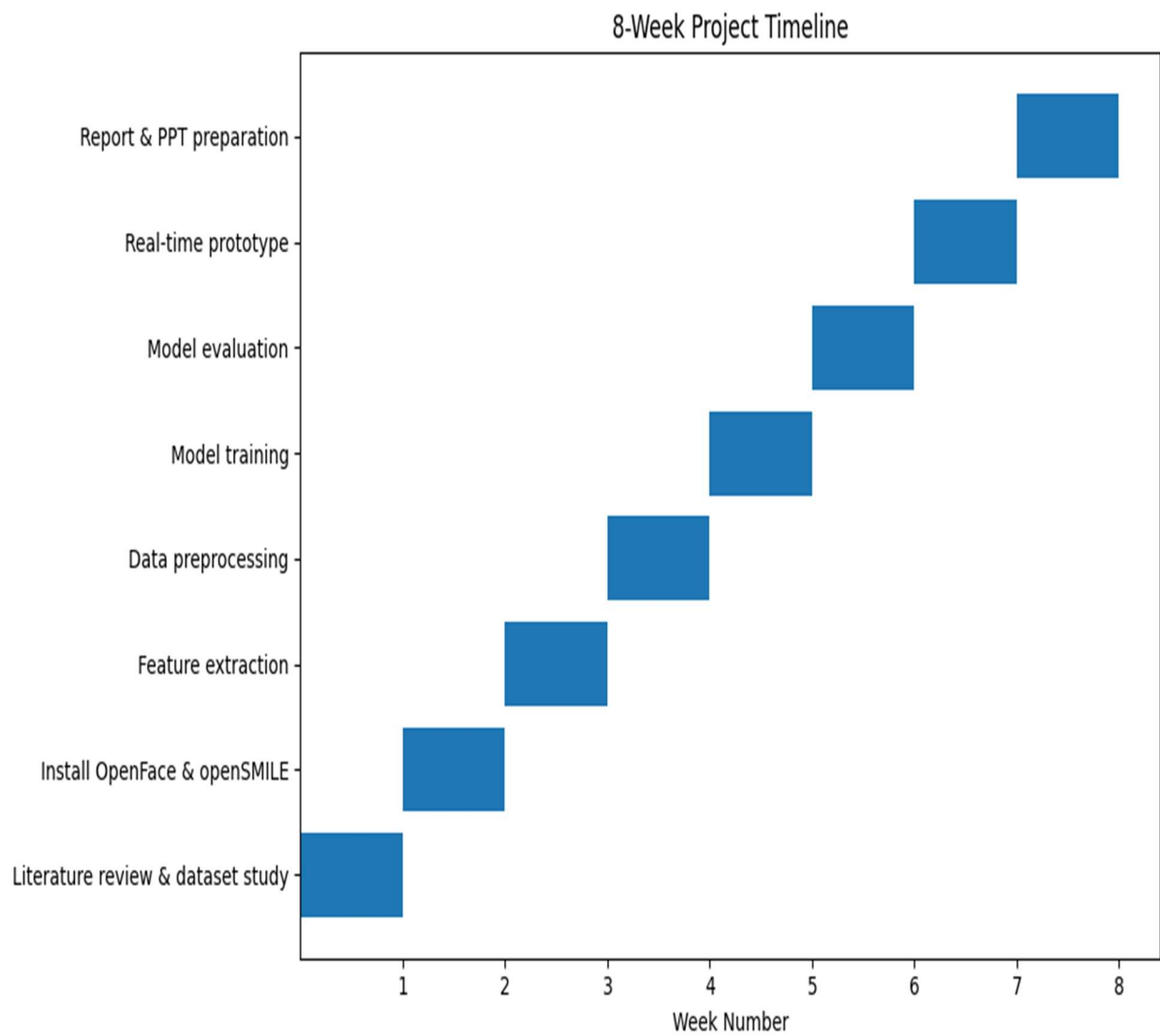
The project will be entirely developed using **Python**, due to its strong support for machine learning, data processing, and computer vision applications. Feature extraction will be performed using two specialized tools: **OpenFace** for facial analysis and **openSMILE** for speech signal processing. Development and experimentation will be carried out using **Jupyter Notebook or Visual Studio Code (VS Code)**, which provide an interactive and structured coding environment.

The system will rely on several Python libraries, including:

- **NumPy** for numerical computations
- **Pandas** for data handling and preprocessing
- **Scikit-learn** for machine learning model training and evaluation
- **OpenCV** for real-time video processing
- **Streamlit or Flask** for building an interactive stress dashboard

These tools together enable seamless integration of real-time data capture, feature extraction, machine learning prediction, and user-friendly visualization.

PROJECT TIMELINE



PROJECT OUTCOME

The primary outcome of this project is the successful design, development, and implementation of a **Real-Time Multimodal Stress Screening System Using Voice and Facial Cues** that integrates artificial intelligence, machine learning, and affective computing techniques. The project demonstrates how behavioral signals can be used for automated stress assessment instead of relying only on subjective questionnaires.

The project delivers a **fully functional real-time prototype** that uses a webcam and microphone to capture live audio and video, extracts features through OpenFace and openSMILE, and classifies stress into **Low, Moderate, and High** using a trained machine learning model. This proves that reliable stress screening can be achieved using affordable hardware and open-source tools.

A major outcome is the **interactive Stress Dashboard**, which visually displays stress level along with a confidence score. This makes the system easy to understand for students, teachers, corporate professionals, and researchers without requiring technical expertise.

From a research perspective, the project produces a **structured technical report** suitable for IEEE-style submission, covering literature review, methodology, system design, experimentation, and evaluation. The system is validated using the **DAIC-WOZ dataset**, and performance is measured using accuracy, precision, recall, and confusion matrix.

Practically, the system supports **early stress awareness and mental health monitoring** in schools, workplaces, hospitals, and defense settings, while remaining an ethical screening tool rather than a medical diagnosis system.

The project also establishes a foundation for **future enhancements**, including wearable sensors, deep learning models, mobile app development, and cloud deployment. Overall, the project successfully delivers an intelligent, real-time, and socially relevant stress screening system.